

<https://doi.org/10.1038/s41545-026-00610-6>

Operational evidence standards for machine learning in wastewater treatment

Siyuan Jiang, Ying Yang, Keqiang Zhang, Yaoru Mao & Xiuwen Cheng

Check for updates

Machine learning studies in water treatment often report high retrospective accuracy but rarely show whether models can support operational decisions. We analyse 423 full-text studies and a bounded six-plant wastewater-treatment scenario. Plant deployment (2.8%), real-time testing (5.2%), future-facing validation (18.2%) and uncertainty reporting (8.5%) remain uncommon. We propose an operation-aware evidence standard linking sensing context, validation timing, uncertainty and control role to deployment claims.

Why operational evidence matters

Machine learning (ML) is increasingly used to predict water-treatment performance, optimise process settings and support monitoring decisions^{1–5}. The reported accuracy of these models is often high. In our 423-study full-text corpus, 202 studies report R^2 as the primary metric and the median reported R^2 is 0.97. This number is informative about fit to historical records, but it does not by itself show whether a model can guide an operating decision whose outcome is not yet known.

The operating question is narrower and more demanding than retrospective fit. A plant operator deciding how much coagulant to dose, whether to investigate a predicted permit exceedance, or whether to allow an ML output into a control loop must act on measurements available before the response is observed. A model can interpolate within a curated dataset while remaining operationally uninformative when sensors are sparse, process states are partly observed, compliance-relevant outputs are delayed or future data leak into validation; benchmark-simulation and digital-twin studies can help define operating settings, but they still require clear reporting of what information is available before a plant decision is made^{6–11}. In regulated wastewater settings, these distinctions matter because ML-assisted actions may affect compliance and safety responsibilities^{12–15}.

Here we examine whether the evidence reported in water-treatment ML matches this operational decision context. We combine two empirical components. First, a 423-study evidence map quantifies how often studies report deployment-relevant evidence practices. Second, a six-plant wastewater-treatment scenario tests a bounded routine-monitoring question under forward-looking validation. We use these results to formulate an operation-aware evidence standard that links model claims to sensing context, validation timing, uncertainty and control role. The six-plant

scenario is used as an illustration of evidence requirements, not as a general benchmark for all water-treatment ML.

What the literature reports

The literature evidence map shows a large gap between accuracy reporting and operation-facing evidence. In the 423-study corpus, plant deployment is reported in only 12 studies (2.8%, Wilson 95% CI 1.6–4.9%), real-time testing with live plant data in 5.2%, and uncertainty quantification in 8.5%. Future-facing validation, defined as temporal split, walk-forward or external validation, appears in 18.2%. Public code (7.3%) and public data (13.2%) are also uncommon, limiting independent retesting of most claims (Fig. 1). The field has grown from 14 papers in 2018 to 104 in 2023, and artificial neural networks appear in 68.6% of studies, but the evidence practices needed for operation remain much less frequently reported.

The low deployment rate is not driven by a single document type or journal subset. Restricting to journal articles gives a 3.2% deployment rate, a core-water-journal subset gives 2.2%, and the high-impact subset contains no deployed studies at all (Tables S11 and S12). All 12 deployment labels in the analysis corpus were checked by full-text manual review. These filters and checks do not establish prevalence beyond the screened corpus, but they show that the operational-evidence gap is not a simple artefact of conference-like records or lower-visibility venues.

We therefore treat the evidence-map indicators as separate descriptive fields rather than as a composite score. Plant deployment is an outcome, not a component of an index used to explain deployment. R^2 -tier deployment counts are reported descriptively, and we do not infer an association between retrospective accuracy and deployment from the small number of deployed studies. Keeping deployment, validation timing, uncertainty, code availability, and data availability separate is what makes the missing evidence visible.

A bounded six-plant operational test

The plant analysis asks a deliberately constrained question: using only routine influent chemistry available before effluent feedback, how well can the next compliance-relevant effluent value be predicted? The dataset contains six anonymised operating wastewater treatment plants (6006 timestamped records, 8 to 13 routinely monitored influent variables per plant and compliance-critical effluent targets including COD, $\text{NH}_3\text{-N}$, TN, TP and TSS where available). Records sharing a timestamp were kept within the same partition, and ridge regression, random forest and histogram gradient boosting were evaluated under random cross-validation and walk-forward evaluation against mean, persistence and lagged-effluent baselines.

Across 29 plant-target tasks, the median feedforward R^2 was 0.063 under random cross-validation and 0.029 under walk-forward evaluation. At the plant level, the best feedforward random-CV mean R^2 was 0.396 and the weakest was -0.098 ; under walk-forward evaluation, five plants

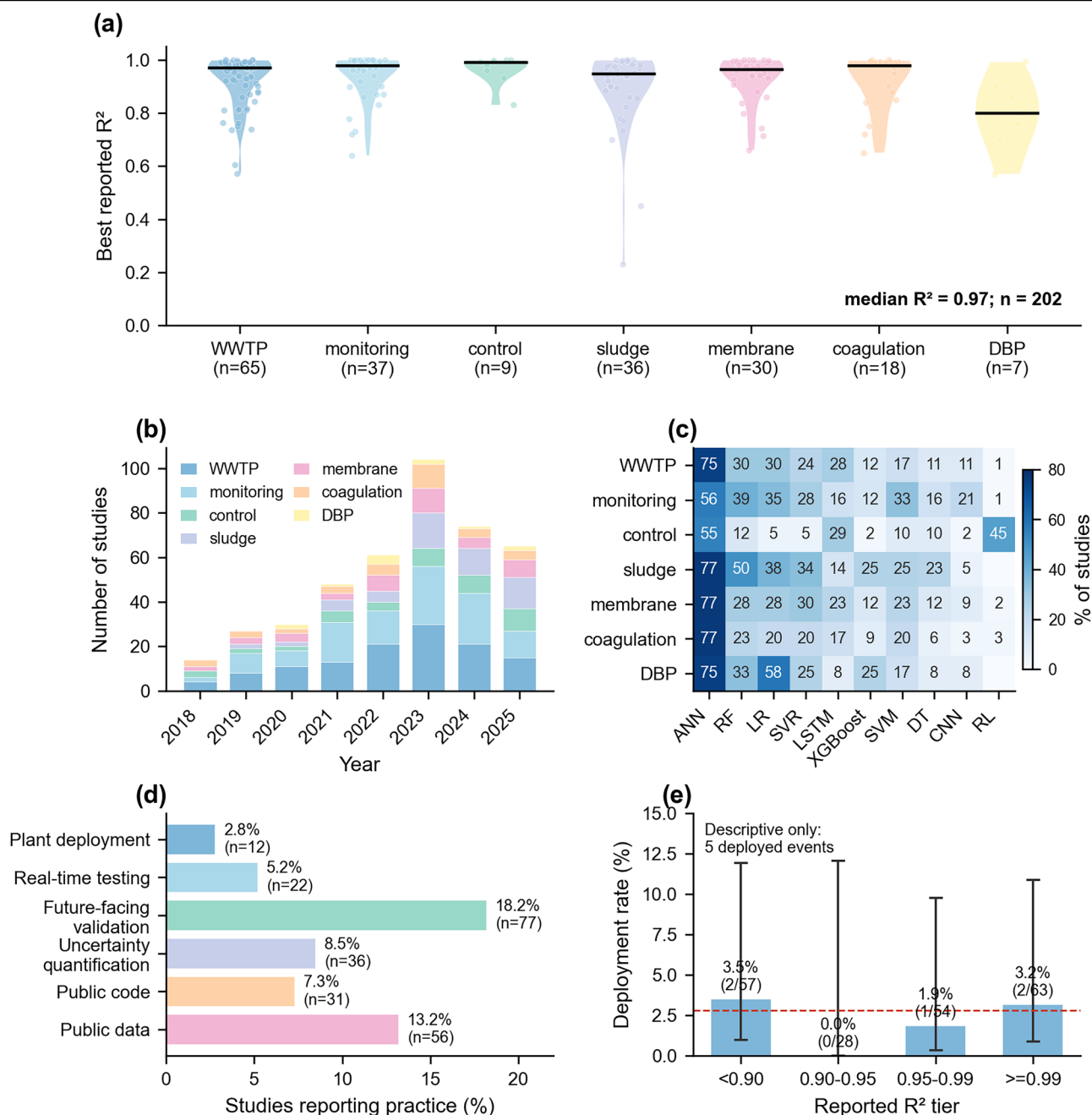


Fig. 1 | Published evidence map for water-treatment ML. **a** Distribution of reported R^2 across seven sub-fields (corpus median $R^2 = 0.97$, $n = 202$). **b** Annual publication trends by sub-field. **c** Algorithm-usage heatmap (percentage of studies per sub-field). **d** Adoption of deployment-relevant evidence practices in the 423-

study corpus. **e** Plant-deployment counts and rates by reported- R^2 tier with Wilson 95% confidence intervals; tier comparisons are descriptive only and not used for association inference. WWTP, wastewater treatment plant; UQ, uncertainty quantification.

remained near zero and the strongest plant mean was 0.099. The median walk-forward R^2 values were -0.024 for the mean baseline, -0.960 for persistence and 0.031 for influent plus lagged effluent (Fig. 2). A daily residence-time lag scan from 0 to 3 days did not raise the influent-only result, and effluent coefficients of variation reached about 105% in one plant, indicating that the weak scores were not simply caused by a nearly constant target (Tables S13-S15).

This result should be interpreted within its stated operating scenario. It does not show that richer sensor suites, mechanistic features, online ammonia or dissolved oxygen histories, or actuator histories would fail in the same way. It shows that when the sensing context is restricted to routine influent chemistry and validation is prospective, the evidence needed for an operational claim is different from the evidence supplied by retrospective fitting alone.

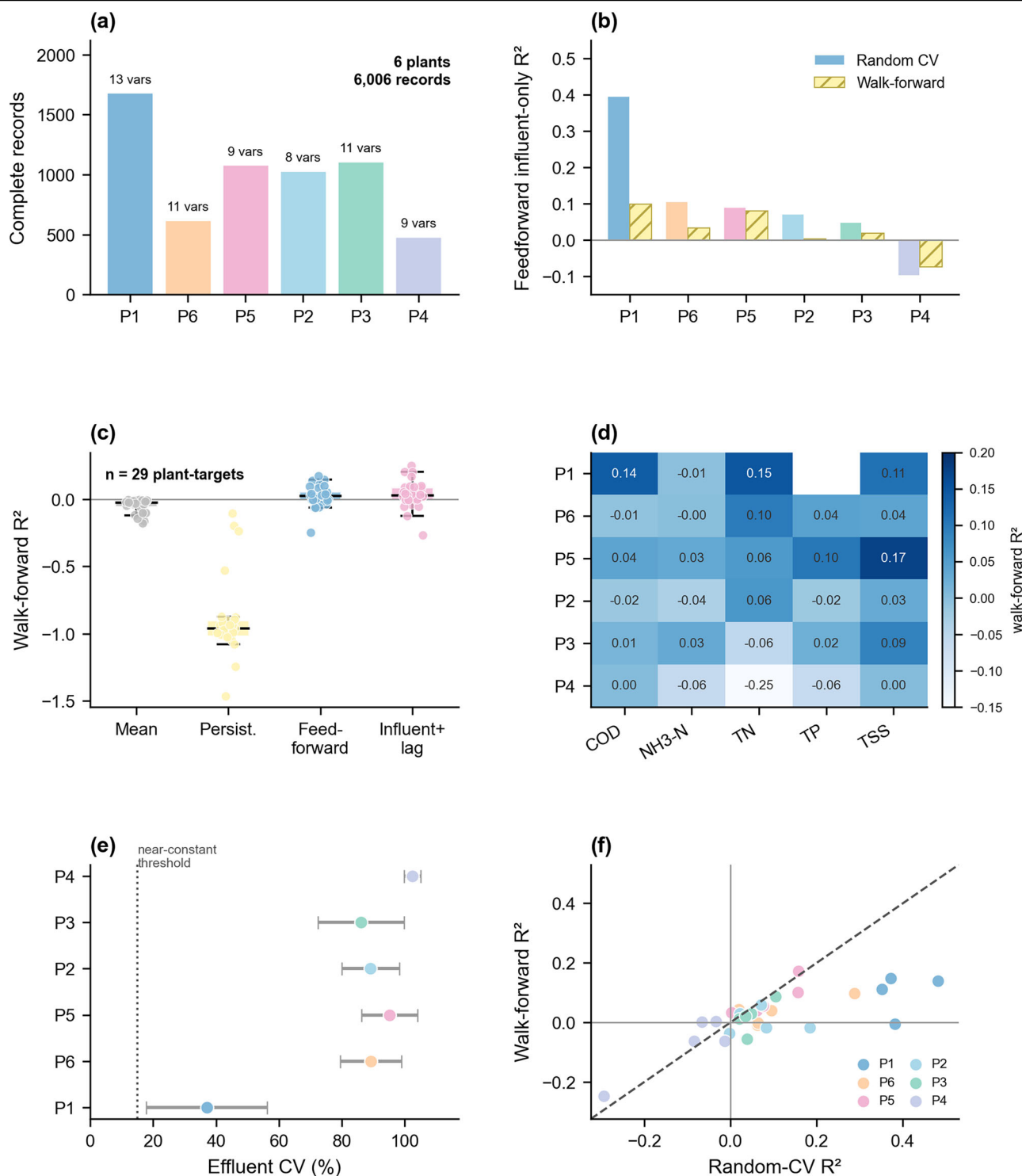


Fig. 2 | Bounded six-plant operational illustration under routine monitoring. Plants are anonymised (Plant 1-6) with relative time. **a** Dataset overview; bars show complete timestamped records and labels the number of routine influent predictors. **b** Feedforward influent-only R^2 by plant under random cross-validation versus date-based walk-forward evaluation. **c** Walk-forward distributions across 29 plant-target

tasks for mean, persistence, feedforward ML and influent-plus-lagged-effluent ML. **d** Per-target feedforward walk-forward R^2 heatmap. **e** Effluent coefficient of variation per plant. **f** Per-target random-CV versus walk-forward R^2 ; points below the diagonal indicate weaker future-facing performance.

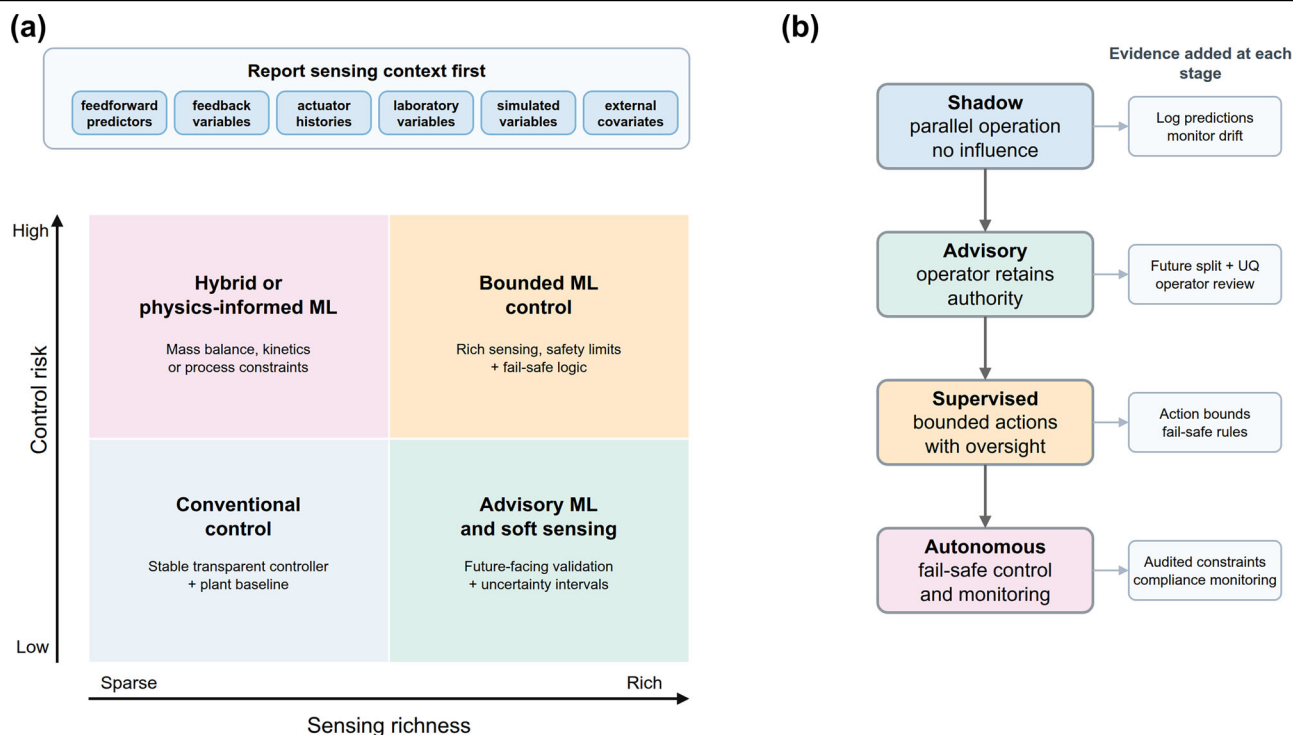


Fig. 3 | Operation-aware evidence standard for water-treatment ML. **a** Reporting first identifies the sensing context and the application suitability that follows from sensing richness and control risk. **b** Staged deployment progresses from shadow to

advisory mode, supervised autonomy and, only where appropriate, bounded autonomous operation, with evidence requirements rising at each stage.

An operation-aware evidence standard

The evidence map and the plant scenario point to the same practical requirement: accuracy metrics need to be accompanied by evidence that matches the intended operating role. We define an operation-aware evidence standard as a minimum set of disclosures that states what a model is for, what information it can see at the decision point, how it was validated in time, how uncertainty is reported, and what happens if the output is acted upon.

The standard has four core axes. Sensing context labels each predictor as a feedforward variable available before the decision, a feedback variable observed after the response, an actuator history, a laboratory measurement, a simulated variable or an external covariate. Validation timing asks whether testing is random, temporal, walk-forward, or external. Uncertainty asks whether prediction intervals, calibration or abstention rules are reported. Control role and risk ask whether the output is used for offline analysis, shadow testing, advisory decisions, supervised autonomy or closed-loop control. For compliance-relevant effluent prediction, the minimum evidence should include a future-facing validation design and uncertainty intervals. For advisory dosing, authors should report reversibility, operator review and comparison with the relevant conventional rule. For closed-loop control, the evidence should include safety constraints, fail-safe logic and comparison with a conventional or mechanistic baseline^{16–20}.

The same axes support staged deployment, recognising practical constraints on AI use in wastewater treatment²¹. In shadow mode a model runs beside existing operations without affecting decisions; in advisory mode operators review predictions and retain authority; in supervised autonomy the model can execute bounded actions under oversight and fail-

safe rules; and fully autonomous operation should be reserved for cases with future-facing validation, uncertainty calibration, operational monitoring and documented control constraints^{22,23}. For weakly observed or high-risk processes, advisory deployment may be the appropriate endpoint rather than a transition to full autonomy (Fig. 3).

Implications and limitations

This study reframes the evidence problem in water-treatment ML. The issue is not that retrospective accuracy is useless; it is that retrospective accuracy answers a different question from operational use. The literature evidence map shows that deployment-relevant reporting remains uncommon, while the plant scenario shows how a model claim changes when sensing context and validation timing are made explicit. Together, these results support a reporting standard rather than a single universal performance threshold.

For authors, the standard provides a way to align model claims with the evidence actually collected. For utilities, it separates low-risk shadow or advisory use from higher-risk control applications. For reviewers and regulators, it identifies the disclosures needed before an ML-assisted decision affects compliance-relevant operations. Physics-informed and hybrid ML can fit naturally into this standard when physical constraints are tested and reported as evidence, rather than used only as labels; in our corpus, only 47 of 423 studies (11.1%) mention such concepts.

Several limitations bound the conclusions. The 423-study corpus is a screened full-text corpus from 2018 to 2025, not a census of every water-treatment ML paper. The extraction workflow used LLM-assisted coding followed by validation and manual checks, so rare or ambiguous reporting practices may still be missed. Deployment counts are small and are used

descriptively only. The six-plant records are anonymised and raw data cannot be released because of utility confidentiality; public verification is therefore based on derived outputs. The plant scenario tests one routine-monitoring feedforward task and should not be generalised to richer instrumentation, different targets or mechanistically constrained control designs.

The constructive implication is straightforward. Studies should report the intended operating role, the sensors available at decision time, the validation design, uncertainty behaviour and control interface. Utilities can use shadow and advisory modes to test models without surrendering authority. The field does not need to replace accuracy metrics; it needs to pair them with evidence that matches the decision being claimed.

How the evidence was assembled

Literature evidence map. The literature analysis used a 500-record source extraction pool and a 423-study full-text analysis corpus covering 2018–2025. Records were screened for water-treatment ML relevance and coded for sub-field, model family, data source, validation method, reported performance metric, deployment evidence, real-time testing, uncertainty quantification, interpretability, code availability and data availability. The primary manuscript numbers are taken from the project truth tables generated from these coded records.

Extraction reliability was evaluated against a 50-paper gold-standard subset. In that benchmark, boolean-field mean accuracy was 0.973, categorical-field mean accuracy was 0.89 and macro F1 was 0.849. Deployment labels were treated as load-bearing fields: all positive deployment labels in the 423-study corpus were checked by full-text manual review. Robustness analyses repeated the main deployment and reporting summaries after restricting to journal articles, excluding conference-like records, restricting to core water journals and considering a high-impact-journal subset.

Six-plant operational scenario. The plant analysis used derived records from six anonymised wastewater treatment plants. Plants were labelled 1–6, real dates were replaced by relative time and raw records were not released. The modelling table contained 6006 complete timestamped records. Each plant had 8–13 routinely monitored influent variables and up to five compliance-relevant effluent targets, depending on availability. The feedforward scenario used only routine influent chemistry available before effluent feedback was observed.

For each plant-target task, ridge regression, random forest, and histogram gradient boosting were compared with mean, persistence, and influent-plus-lagged-effluent baselines. Random cross-validation and walk-forward evaluation used the same date-based split logic so that records sharing a timestamp could not straddle training and test partitions. Walk-forward folds retrained on past date blocks with fixed hyperparameters and no tuning on the future test block. Model skill was summarised as R^2 , with negative values retained because they indicate performance worse than the relevant baseline.

Supporting Information reports the screening, extraction, reliability and modelling details, including the robustness summaries in Tables S11 and S12, the six-plant residence-time and variability checks in Tables S13–S15, and the staged evidence framework in Table S16. Public data and code are provided through Zenodo as described below.

Data availability

The literature corpus, manual-review records, extraction prompts, LLM benchmark summaries, figure source data and anonymised six-plant

derived outputs are available at Zenodo: <https://doi.org/10.5281/zenodo.20604112>.

Code availability

All analysis, validation and figure-generation scripts needed to reproduce the public results are available in the Zenodo package at <https://doi.org/10.5281/zenodo.20604112> and in the project repository. The released scripts include corpus cleaning, extraction reliability checks, manual-review summaries, six-plant derived-output analysis, figure generation, manuscript validation and submission-package preparation.

Siyuan Jiang, Ying Yang, Keqiang Zhang, Yaoru Mao & Xiuwen Cheng✉

Key Laboratory for Environmental Pollution Prediction and Control, Gansu Province, College of Earth and Environmental Sciences, Lanzhou University, Lanzhou, China. ✉e-mail: chengxw@lzu.edu.cn

Received: 9 June 2026; Accepted: 10 July 2026;

Published online: 17 July 2026

References

- Newhart, K. B. et al. Data-driven performance analyses of wastewater treatment plants: a review. *Water Res.* **157**, 498–513 (2019).
- Alvi, M. et al. Deep learning in wastewater treatment: a critical review. *Water Res.* **245**, 120518 (2023).
- Zhi, W., Appling, A. P., Golden, H. E., Podgorski, J. & Li, L. Deep learning for water quality. *Nat. Water*. **2**, 228–241 (2024).
- Safeer, S. et al. A review of artificial intelligence in water purification and wastewater treatment: recent advancements. *J. Water. Process Eng.* **49**, 102974 (2022).
- Huang, R., Ma, C., Ma, J., Huangfu, X. & He, Q. Machine learning in natural and engineered water systems. *Water Res.* **205**, 117666 (2021).
- Quiñero-Candela, J., Sugiyama, M., Schwaighofer, A. & Lawrence, N. D. Dataset Shift in Machine Learning (MIT Press, 2009).
- Alex, J. et al. Benchmark Simulation Model no.1 BSM1 (Lund University, 2008).
- Paleyes, A., Urma, R.-G. & Lawrence, N. D. Challenges in deploying machine learning: a survey of case studies. *ACM Comput. Surv.* **55**, 1–29 (2022).
- Wang, A. J. et al. Digital twins for wastewater treatment: a technical review. *Engineering* **36**, 21–35 (2024).
- Rudin, C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nat. Mach. Intell.* **1**, 206–215 (2019).
- Roberts, D. R. et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* **40**, 913–929 (2017).
- Federal Water Pollution Control Act, 33 U.S.C. §1251 et seq. (1972).
- Council Directive 91/271/EEC of 21 May 1991 concerning urban waste water treatment. Off. J. Eur. Communities L135, 40–52 (1991).
- Directive (EU) 2024/3019 of the European Parliament and of the Council of 27 November 2024 concerning urban wastewater treatment (recast). Off. J. Eur. Union L 2024/3019 (2024).
- Amodei, D. et al. Concrete problems in AI safety. Preprint at <https://doi.org/10.48550/arXiv.1606.06565> (2016).
- Bratby, J. Coagulation and Flocculation in Water and Wastewater Treatment (IWA Publishing, 2006).
- Sutton, R. S. & Barto, A. G. Reinforcement learning: an introduction 2nd edn (MIT Press, 2018).
- Camacho, E. F. & Bordons, C. Model Predictive Control 2nd edn (Springer, 2007).
- Salvato, E., Fenu, G., Medvet, E. & Pellegrino, F. A. Crossing the reality gap: a survey on sim-to-real transferability of robot controllers in reinforcement learning. *IEEE Access* **9**, 153171–153187 (2021).
- Metcalfe & Eddy, AECOM. Wastewater Engineering: Treatment and Resource Recovery 5th edn (McGraw-Hill Education, 2014).
- Capodaglio, A. G. & Callegari, A. Use, potential, needs, and limits of AI in wastewater treatment applications. *Water*. **17**, 170. <https://doi.org/10.3390/w17020170>
- Mankins, J. C. Technology Readiness Levels: A White Paper (NASA Office of Space Access and Technology, 1995).
- SAE International. Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles J3016_202104 (SAE International, 2021). https://doi.org/10.4271/J3016_202104

Acknowledgements

This study was supported by the Key Research and Development Program Project of Gansu Province (25YFFA003), Youth Science and Technology Talent Innovation Project of Lanzhou City (2023-QN-80), Gansu Provincial Youth Science and Technology Fund Project (26JRR223) and the Fundamental Research Funds for the Central Universities (lzuibky-2023-ey14, lzuibky-2025-pd09).

Author contributions

S.J. conceived the study, curated the literature corpus, performed the analysis, generated figures and drafted the manuscript. Y.Y. contributed to study design, literature interpretation and manuscript revision. K.Z. contributed to data curation, validation checks and interpretation of the operational evidence framework. Y.M. contributed to literature screening, data checking and manuscript review. X.C. supervised the study, secured funding, guided interpretation and revised the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41545-026-00610-6>.

Correspondence and requests for materials should be addressed to Xiuwen Cheng.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026